DNS SERVER CONFIGURATION – MUSICMANIA.IN

Practical Assignment — Questions and Answers

1. Install and Set Up a DNS Server and Create a Zone

Question 1: Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

Answer

A Domain Name System (DNS) server translates human-readable domain names into IP addresses. In this assignment, a local DNS server will host a private practice zone named musicmania.in. The zone is for lab use; it does not make the domain publicly available on the Internet.

Option A — BIND on Linux (Ubuntu/Debian)

1. Create or start a Linux virtual machine and make sure it has a stable lab IP address. This example uses 192.168.1.10 for the DNS server.
2. Update package information and install BIND9 and its utilities:

```
sudo apt update
sudo apt install bind9 bind9utils dnsutils
```

3. Check that the DNS service is running:

```
sudo systemctl status bind9
```

4. Open the local BIND configuration file:

```
sudo nano /etc/bind/named.conf.local
```

5. Add a master zone declaration:

```
zone "musicmania.in" {
   type master;
   file "/etc/bind/db.musicmania.in";
};
```

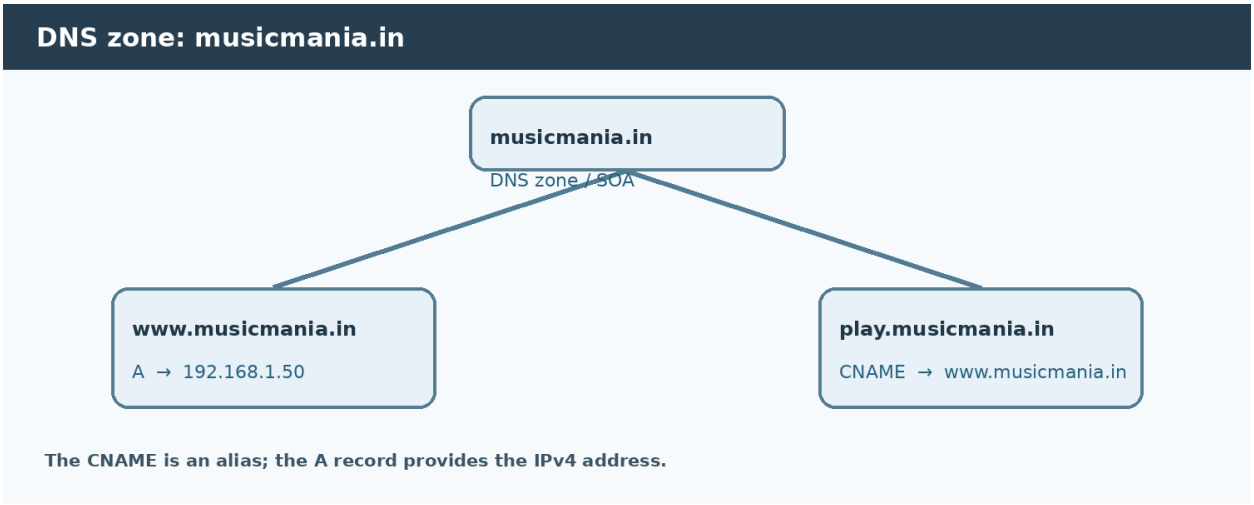


Figure 1. Logical DNS zone structure for musicmania.in.

1. Install and Set Up a DNS Server — Zone File and Windows Alternative

Create the zone file

Create the zone file using the path specified in the zone declaration:

sudo nano /etc/bind/db.musicmania.in

Enter the following example zone data. Replace the administrator email-style field if required by your lab policy.

```
$TTL 3600
@   IN  SOA ns1.musicmania.in. admin.musicmania.in. (
      2026110601 ; Serial (increase after every edit)
      3600       ; Refresh
      900        ; Retry
      604800     ; Expire
      3600       ; Negative cache TTL
)
@       IN  NS      ns1.musicmania.in.
ns1     IN  A       192.168.1.10
```

Validate and restart BIND

```
sudo named-checkconf
sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
sudo systemctl restart bind9
```

If the configuration check reports no errors and the zone check reports OK, the zone file syntax is valid. The DNS server should now be able to serve the zone to clients that query it.

Option B — Simple DNS Plus on Windows

- Install Simple DNS Plus on a Windows machine or VM and open its management interface.
- Create a new primary/authoritative zone named musicmania.in.
- Add the required DNS records in the zone editor and apply/save the configuration.
- Allow DNS queries through the Windows firewall for the lab network, and configure the client to use the server's IP address as its DNS server.

Note: Menu names can vary by software version. The BIND commands below are used for the remaining examples.

2. Add an A Record and Verify It with nslookup or dig

Question 2: Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works.

Answer

An A (Address) record maps a hostname to an IPv4 address. Add the following line to /etc/bind/db.musicmania.in:

www IN A 192.168.1.50

After editing the zone file, increase the SOA serial number so secondary DNS servers and caches can recognize the updated zone version. Save the file.

Check and reload the DNS service

sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
sudo systemctl reload bind9

Verify using dig

dig @127.0.0.1 www.musicmania.in A

The @127.0.0.1 part tells dig to query the DNS server running on the same machine. Look for an ANSWER SECTION containing www.musicmania.in and 192.168.1.50.

Verify using nslookup

nslookup www.musicmania.in 127.0.0.1



Figure 2. Sample DNS lookup output. It is illustrative; your actual response depends on your configuration.

Record	Type	Value	Meaning
www.musicmania.in	A	192.168.1.50	Maps the www host to an IPv4 address.
musicmania.in	SOA	ns1.musicmania.in / admin contact	Identifies zone authority and timing values.
musicmania.in	NS	ns1.musicmania.in	Names the authoritative name server.

Conclusion: The A record is working when a query sent directly to the local DNS server returns 192.168.1.50 for www.musicmania.in.

3. Create a CNAME Record and Test the Alias

Question 3: Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.

Answer

A CNAME (Canonical Name) record makes one DNS name an alias of another DNS name. In this task, play.musicmania.in will be an alias for www.musicmania.in. The target hostname then resolves through its A record to 192.168.1.50.

Add the CNAME record

Add this line to the same BIND zone file:

```
play    IN  CNAME   www.musicmania.in.
```

The final dot in www.musicmania.in. is important in a BIND zone file because it marks the name as fully qualified. A CNAME owner name should not have other ordinary data records at the same name.

Validate and reload

```
sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
sudo systemctl reload bind9
```

Test the alias

```
dig @127.0.0.1 play.musicmania.in A
nslookup play.musicmania.in 127.0.0.1
```

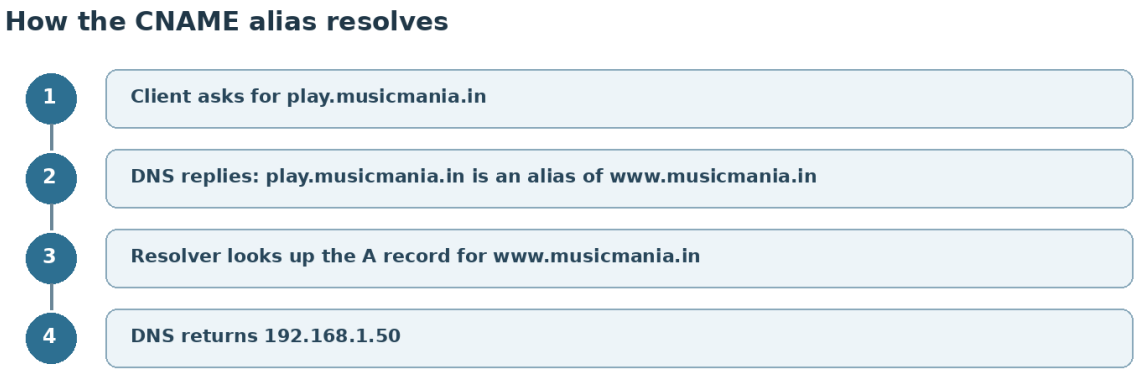


Figure 3. Resolution path: the alias points to www.musicmania.in, whose A record returns 192.168.1.50.

Expected result

The DNS answer should identify play.musicmania.in as a CNAME for www.musicmania.in and ultimately provide the A-record address 192.168.1.50. The alias does not contain a separate IP address; it points to the canonical hostname.

Conclusion: The CNAME record allows an alternate hostname to resolve through the existing www A record.

4. Troubleshoot DNS When musicmania.in Does Not Resolve

Question 4: Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig. Include at least three different troubleshooting commands or checks.

Answer

DNS failures can occur because of incorrect zone syntax, missing records, an outdated SOA serial number, a stopped service, firewall rules or a client using the wrong DNS server. The following process checks the most common causes in a logical order.

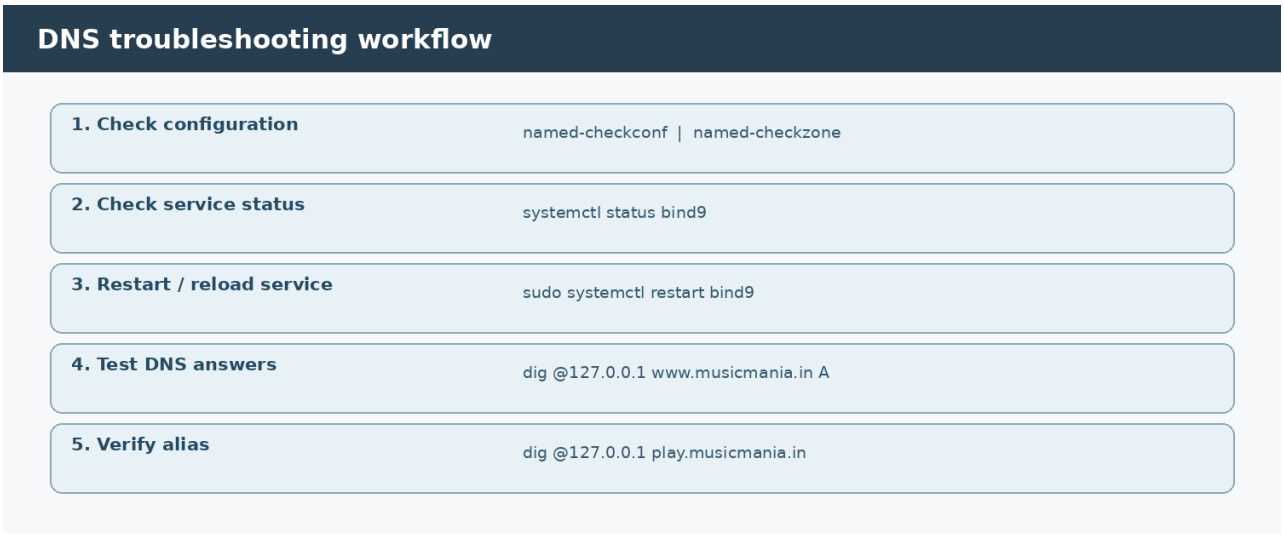


Figure 4. DNS troubleshooting workflow with key commands.

Step 1 — Check configuration syntax

sudo named-checkconf

This checks BIND's main configuration files for syntax errors. If an error is reported, inspect the file and correct the indicated line before restarting.

Step 2 — Validate the zone file

sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in

This checks the zone file for record and formatting errors. Confirm that the A and CNAME records are spelled correctly and that the CNAME target ends with a dot.

Step 3 — Check service status

sudo systemctl status bind9

Confirm that the service is active. If it is stopped or failed, inspect the reported error details.

4. DNS Troubleshooting — Continued

Step 4 — Restart or reload the DNS service

sudo systemctl restart bind9
sudo systemctl status bind9

Restarting applies the corrected configuration. If only zone data changed and the configuration is valid, a reload can be used instead:

sudo systemctl reload bind9

Step 5 — Inspect service logs

sudo journalctl -u bind9 --since "10 minutes ago"

Review recent logs for zone-loading errors, permission problems, port conflicts or configuration failures.

Step 6 — Test directly against the DNS server

dig @127.0.0.1 musicmania.in SOA
dig @127.0.0.1 www.musicmania.in A
dig @127.0.0.1 play.musicmania.in A

These tests query the local server directly and help distinguish server/zone problems from client DNS configuration problems.

Step 7 — Check the client DNS settings

On Linux, check the resolver configuration using resolvectl status or cat /etc/resolv.conf. On Windows, use ipconfig /all. Ensure the client is configured to use the DNS server's lab IP address (for example, 192.168.1.10).

Step 8 — Check network access and firewall

Confirm that the client can reach the DNS server and that DNS traffic is allowed. DNS commonly uses UDP port 53 and may also use TCP port 53, so both should be considered in firewall rules.

Troubleshooting command summary

Command/check	Purpose
named-checkconf	Checks BIND configuration syntax.
named-checkzone musicmania.in /etc/bind/db.musicmania.in	Validates the zone file.
systemctl status bind9	Checks whether the DNS service is running.
journalctl -u bind9	Displays service logs and errors.
dig @127.0.0.1 www.musicmania.in A	Tests the A record directly.
nslookup www.musicmania.in 192.168.1.10	Tests through the lab DNS server.

Final Results, Conclusion and Verification Checklist

Expected DNS records

Name	Record type	Expected value
www.musicmania.in	A	192.168.1.50
play.musicmania.in	CNAME	www.musicmania.in.
ns1.musicmania.in	A	192.168.1.10
musicmania.in	NS	ns1.musicmania.in.

Overall conclusion

This assignment demonstrates how a DNS server can host a local zone and resolve names to IP addresses. The A record maps www.musicmania.in to 192.168.1.50, while the CNAME record makes play.musicmania.in an alias of the www hostname. Configuration validation, service checks, logs and DNS query tools help identify and correct resolution problems.

Screenshots to include in the practical submission

- Task 1: BIND zone declaration and zone file, or the Simple DNS Plus zone creation screen.
- Task 2: Terminal output from dig or nslookup showing www.musicmania.in resolving to 192.168.1.50.
- Task 3: Terminal output showing play.musicmania.in as a CNAME and resolving to 192.168.1.50.
- Task 4: Output of named-checkconf, named-checkzone, systemctl status bind9 and a successful or failed dig query during troubleshooting.

The diagrams and terminal output in this document are illustrative teaching aids, not screenshots from an actual machine. Replace them with your own screenshots after completing the configuration. The example uses a private lab DNS server at 192.168.1.10; ensure that your VM, client and network use matching addresses.